Building Data Apps In Python With Streamlit

Session 1: Setup & Basics

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Exercise: What is a Data App?

- Introduce yourself to your neighbor.
- Ask them 2 questions:
  - *What do you already know about Data Apps, Python and Streamlit?*
  - *What do you want to build after completing this course?*

About me

Ari Lamstein

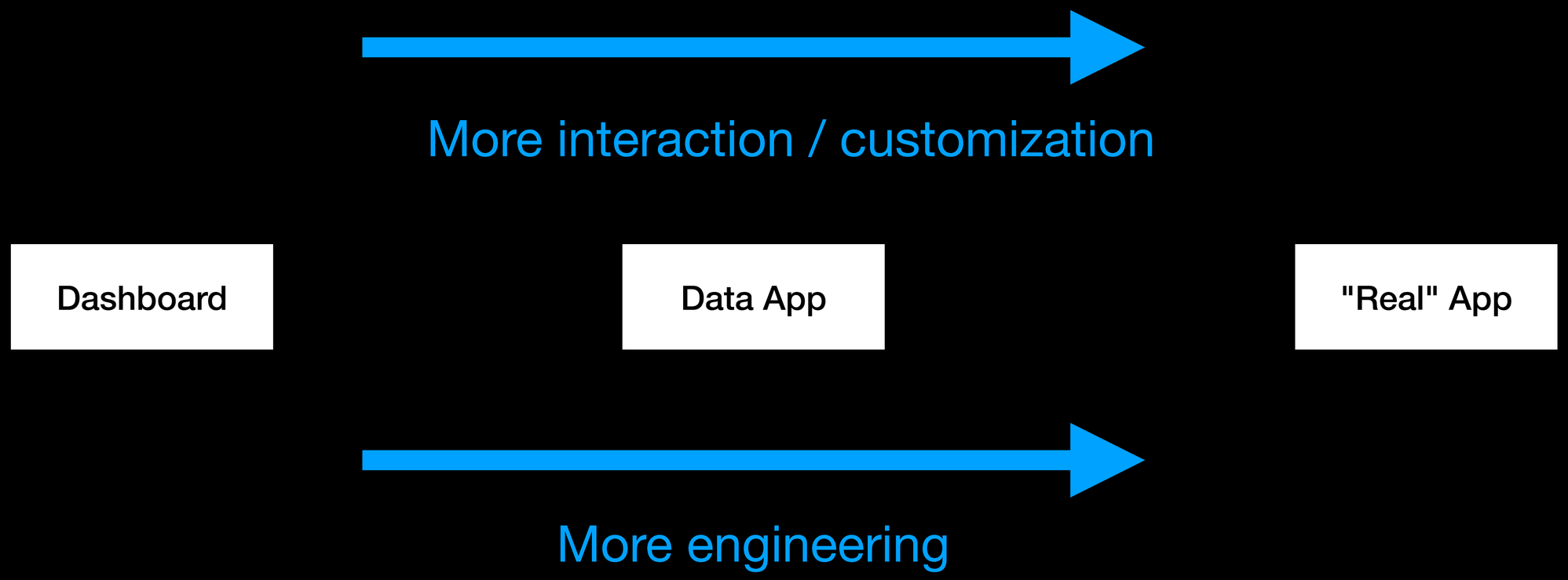
- Started out as a PhD student in Computer Science -> left to join industry
- Worked as a software engineer at large tech companies in San Francisco
- Authored several R packages that deal with analyzing public datasets. R Trainer.
- Recent years: Python.
- Created two data apps using Streamlit:
  - Covid Demographics Explorer
  - Immigration Enforcement Explorer

Exercise: What is a Data App?

- Introduce yourself to your neighbor.
- Ask them 2 questions:
  - *What do you already know about Data Apps, Python and Streamlit?*
    - Streamlit is really popular: downloaded 16M times last month
  - *What do you want to build after completing this course?*
    - I use Streamlit to build data apps that make it easy to understand public / government datasets. This is the course I wish I had when I was starting out!

Exercise: What is a Data App?

- My answer has two parts
  - "Data" apps vs."Real" apps vs. Dashboards
  - Data apps vs. Jupyter Notebooks

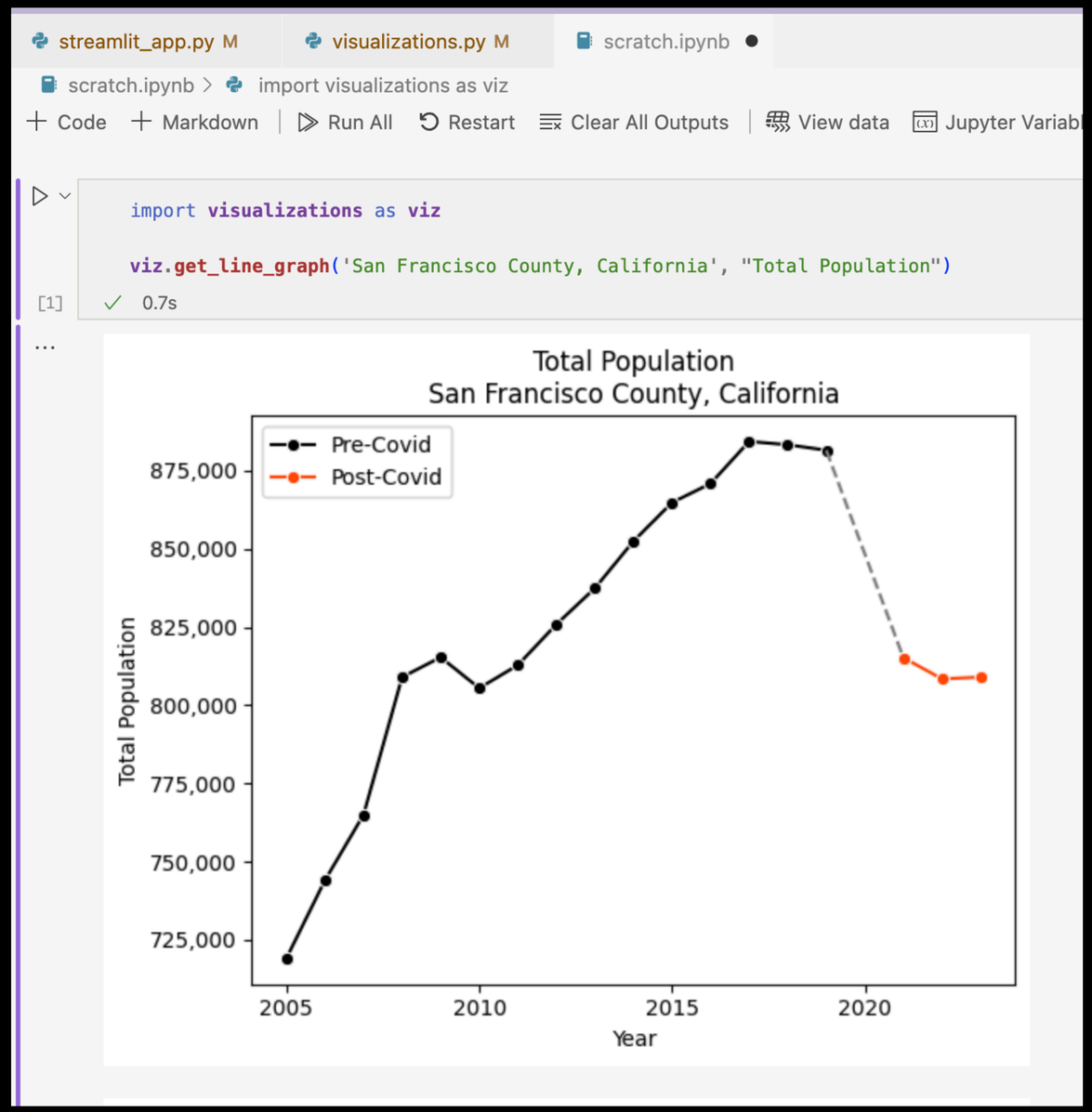

More interaction / customization
Dashboard
Data App
"Real" App
More engineering

Data Apps vs. Jupyter Notebooks

- Jupyter Notebooks let me create a single story about a dataset. I can share the notebook with others for free via github.
- Streamlit lets me create an app to explore a dataset. I can share the app with others for free via Streamlit Cloud.
- I often start with a notebook, and end with a Streamlit app.

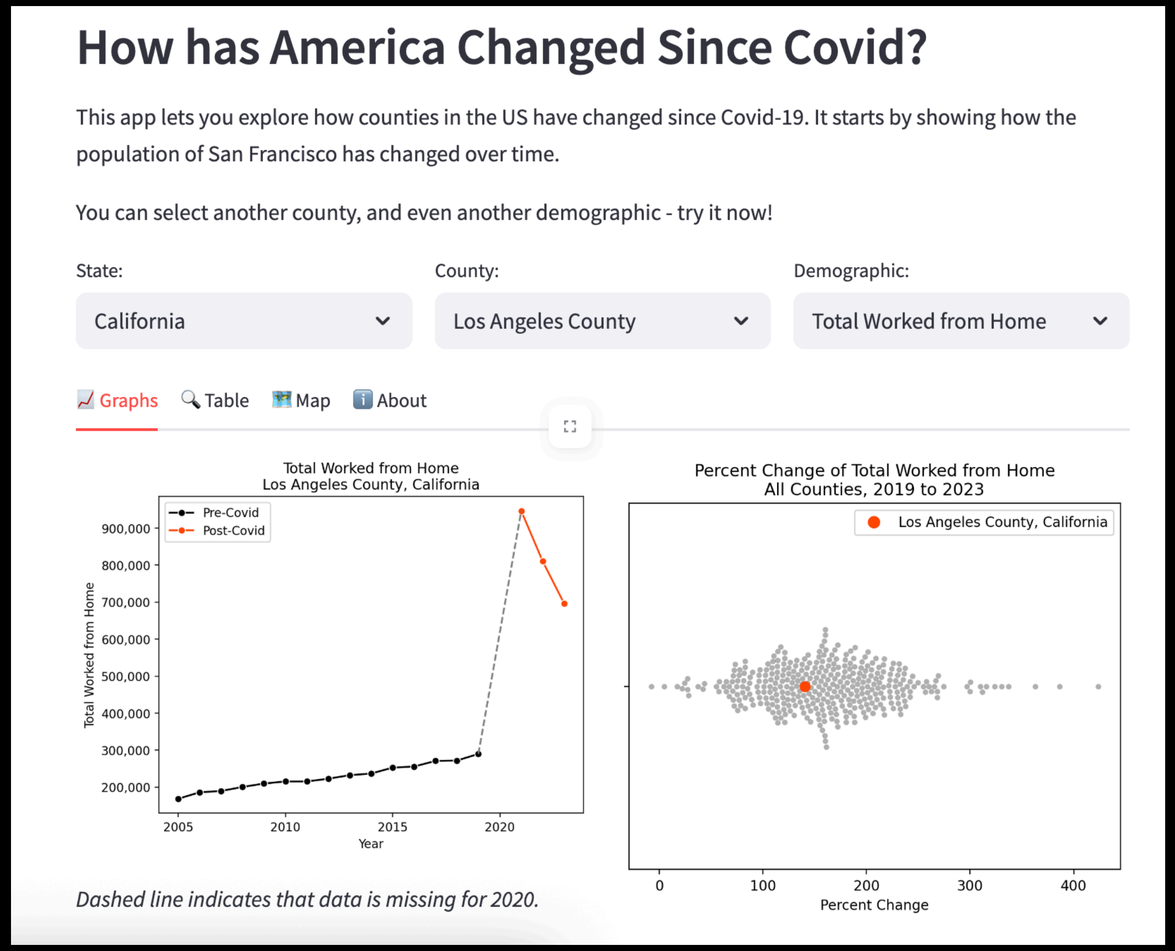
Example Jupyter Notebook

- This notebook tells a story about one city.



Example Streamlit App

- This Streamlit App lets users see how the population of their county changed
- https://census-explorer.streamlit.app/



Course Agenda
(Big Picture)

1. Session 1: Setup & Basics
2. Session 2: "The Loop": Inputs & Graphics
3. Session 3: Organization: UI & Code
4. Session 4: Deployment

1-intro-setup/1-handout.pdf

Get Set Up Before Monday's Workshop 🚀

We'll walk through setup together at the start of the seminar, but if you'd like to hit the ground running, here's how to get everything installed ahead of time:

✅ Step 1: Clone the course repo

Type **git clone https://github.com/arilamstein/streamlit-workshop.git**

✅ Step 2: Create your virtual environment

1. Install uv by following the instructions here: https://docs.astral.sh/uv/getting-started/installation/
2. Restart your terminal after installation.
3. In the course repo, type **uv sync**

✅ Step 3: Verify your setup

Activate the virtual environment:

- Mac/Linux: type **source .venv/bin/activate**
- Windows (Command Prompt): type **.venv\Scripts\activate**

Then launch the demo app:

1. Type **cd 1-intro-setup**
2. Type **streamlit run streamlit_app.py**

A browser should open up showing a table of data - if it does, you're good to go!

🛠️ Editing code and notebooks

You'll need a way to edit Python files and Jupyter notebooks. I recommend VS Code with the "Jupyter" plugin. To test it:

- Open **scratch.ipynb** in the **1-intro-setup** directory.

If you are not using an IDE that supports Jupyter, you can use Jupyter Notebook directly by typing **uv run jupyter notebook scratch.ipynb**

These instructions are also in the repo, but I wanted to post them here for visibility. Feel free to reply if you run into any issues-we'll troubleshoot together on Monday!

Session 1: Setup

1. One-Time Setup
    1. Clone course repo

Clone course repo

- The course repo is at URL is at https://github.com/arilamstein/streamlit-workshop

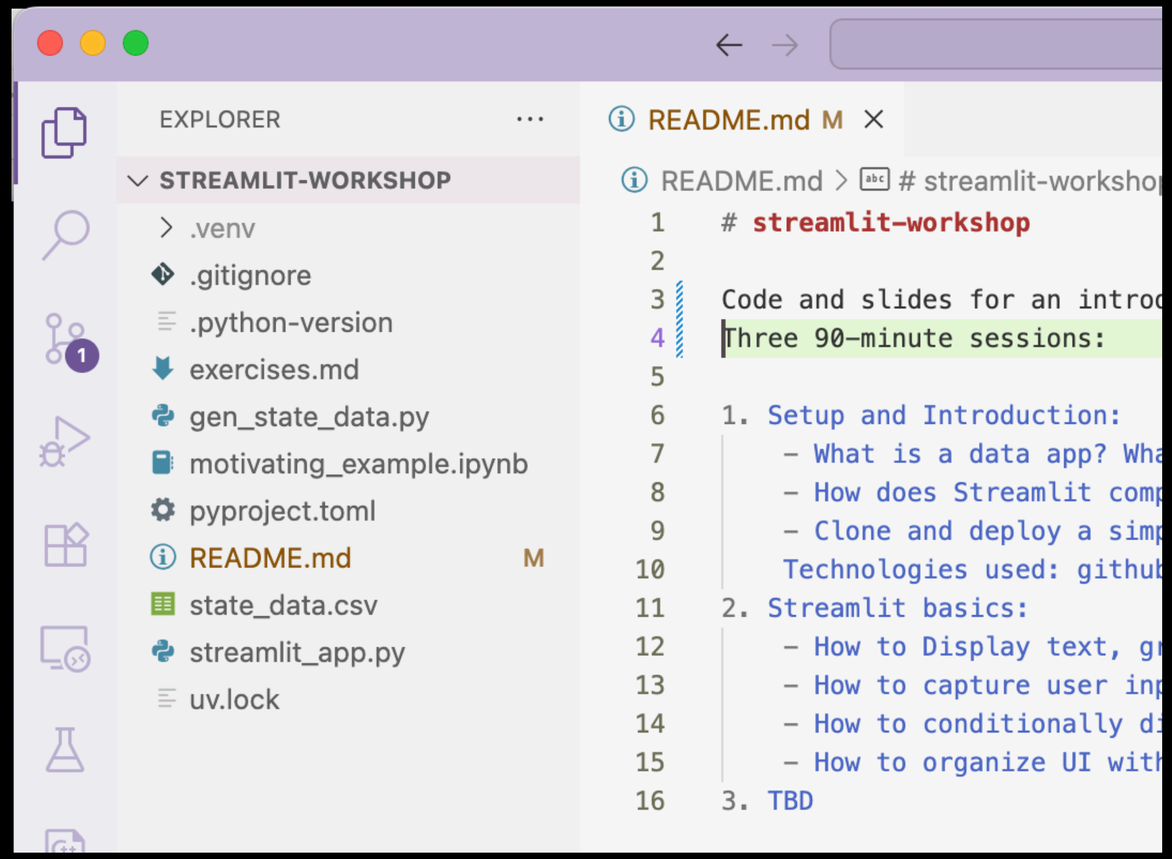
- To clone it, go to the terminal and type

  git clone https://github.com/arilamstein/streamlit-workshop.git

```
➜  test git clone https://github.com/arilamstein/streamlit-workshop.
Cloning into 'streamlit-workshop'...
remote: Enumerating objects: 12, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 12 (delta 0), reused 12 (delta 0), pack-reused 0 (from
Receiving objects: 100% (12/12), 175.62 KiB | 1.66 MiB/s, done.
➜  test ls
streamlit-workshop
```

Open Course Repo

- Open the course repo using your favorite IDE



Session 1: Setup

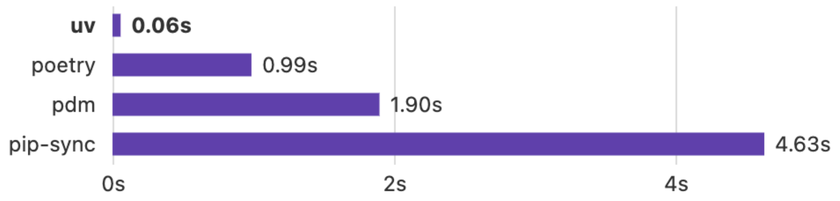
1. One-Time Setup
    1. Clone course repo
    2. Create the virtual environment

uv

- "An extremely fast Python package and project manager, written in Rust"
- Uses 3 files:
  - pyproject.toml
  - uv.lock
  - .python-version

uv

An extremely fast Python package and project manager, written in Rust.



Installing Trio's dependencies with a warm cache.

Highlights

- 🚀 A single tool to replace `pip`, `pip-tools`, `pipx`, `poetry`, `pyenv`, `twine`, `virtualen` and more.
- ⚡ 10-100x faster than `pip`.
- 🗂️ Provides comprehensive project management, with a universal lockfile.
- ✅ Runs scripts, with support for inline dependency metadata.
- 🐍 Installs and manages Python versions.
- 🛠️ Runs and installs tools published as Python packages.
- 🔩 Includes a pip-compatible interface for a performance boost with a familiar CLI.

pyproject.toml

- Specifies a (minium) version of Python for the project
- Specifies packages that need to be installed ("dependencies")
- Specifies minimum versions required

pyproject.toml

```toml
[project]
name = "streamlit-workshop"
version = "0.1.0"
description = "Add your descr
readme = "README.md"
requires-python = ">=3.13"
dependencies = [
    "censusdis>=1.4.2",
    "ipykernel>=6.30.1",
    "kaleido>=1.1.0",
    "pandas>=2.3.2",
    "plotly>=6.3.0",
    "streamlit>=1.49.1",
]
```

uv.lock

- Says the exact version of each package to install
- Ensures reproducible environments
- Eliminates a common source of errors

```
[[package]]
name = "streamlit"
version = "1.49.1"
source = { registry = "https://pypi.org/simple" }
dependencies = [
    { name = "altair" },
    { name = "blinker" },
    { name = "cachetools" },
    { name = "click" },
    { name = "gitpython" },
    { name = "numpy" },
    { name = "packaging" },
    { name = "pandas" },
    { name = "pillow" },
    { name = "protobuf" },
    { name = "pyarrow" },
    { name = "pydeck" },
    { name = "requests" },
    { name = "tenacity" },
    { name = "toml" },
    { name = "tornado" },
    { name = "typing-extensions" },
    { name = "watchdog", marker = "sys_platform != 'darwin'" },
]
sdist = { url = "https://files.pythonhosted.org/packages/24/17/c8024
wheels = [
    { url = "https://files.pythonhosted.org/packages/85/9e/146cdef51
]
```

.python-version

- Says which version of Python to install
  - Major and minor version are specified
  - uv will always install the latest patch version
- This eliminates a common source of errors

.python-version

```
3.13
```

Install uv

- Google "install uv".
- The first hit should be this page.
- Copy the "curl" command into your terminal.

Getting started

Installing uv

Installation methods

Install uv with our standalone installers or your package manager

Standalone installer

uv provides a standalone installer to download and install uv:

macOS and Linux Windows

Use `curl` to download the script and execute it with `sh`:

```
$ curl -LsSf https://astral.sh/uv/install.sh | sh
```

Installing uv

- What I see after installation

```
→  streamlit-workshop git:(main) ✗ curl -LsSf https://astral.sh/uv/install.sh | sh
downloading uv 0.8.17 aarch64-apple-darwin
no checksums to verify
installing to /Users/arilamstein/.local/bin
  uv
  uvx
everything's installed!
WARN: The following commands are shadowed by other commands in your PATH: uv uvx
```

Restart your terminal

uv sync

- In the terminal, go back to the project directory
- Type uv sync in the terminal
- Creates a virtual environment in the folder .venv that is "in sync" with the pyproject.toml, uv.lock and .python-version files
  - This directory will have the correct version of Python, and the correct version of each package the project needs

```
→  streamlit-workshop git:(main) ✗ uv sync
Using CPython 3.13.7
Creating virtual environment at: .venv
Resolved 145 packages in 4ms
Installed 139 packages in 385ms
 + adjusttext==0.8
 + affine==2.4.0
 + alabaster==1.0.0
 + altair==5.5.0
 + appnope==0.1.4
 + asttokens==3.0.0
 + attrs==25.3.0
 + babel==2.17.0
 + beautifulsoup4==4.13.5
 + black==25.1.0
 + bleach==6.2.0
 + blinker==1.9.0
 + cachetools==6.2.0
 + censusdis==1.4.2
 + certifi==2025.1.31
```

Session 1: Setup

1. One-Time Setup
    1. Clone course repo
    2. Create the virtual environment
2. Per-Session Setup
    1. Activate the virtual environment
    2. Run the example Streamlit app

Activate the Virtual Environment

- Activate the virtual environment:
  - **Mac/Linux**: source .venv/bin/activate
  - **Windows (Command Prompt)**: .venv\Scripts\activate

Test the Virtual Environment

- Check that the virtual environment is running:
  - Type **python check_env.py**

```
(streamlit-workshop) ➜  streamlit-workshop git:(main) ✗ python check_env.py
Python executable: /Users/arilamstein/uv-projects/streamlit-workshop/.venv/bin/python
Virtual env active: True
(streamlit-workshop) ➜  streamlit-workshop git:(main) ✗
```

Session 1: Setup

1. One-Time Setup
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Run the Demo App

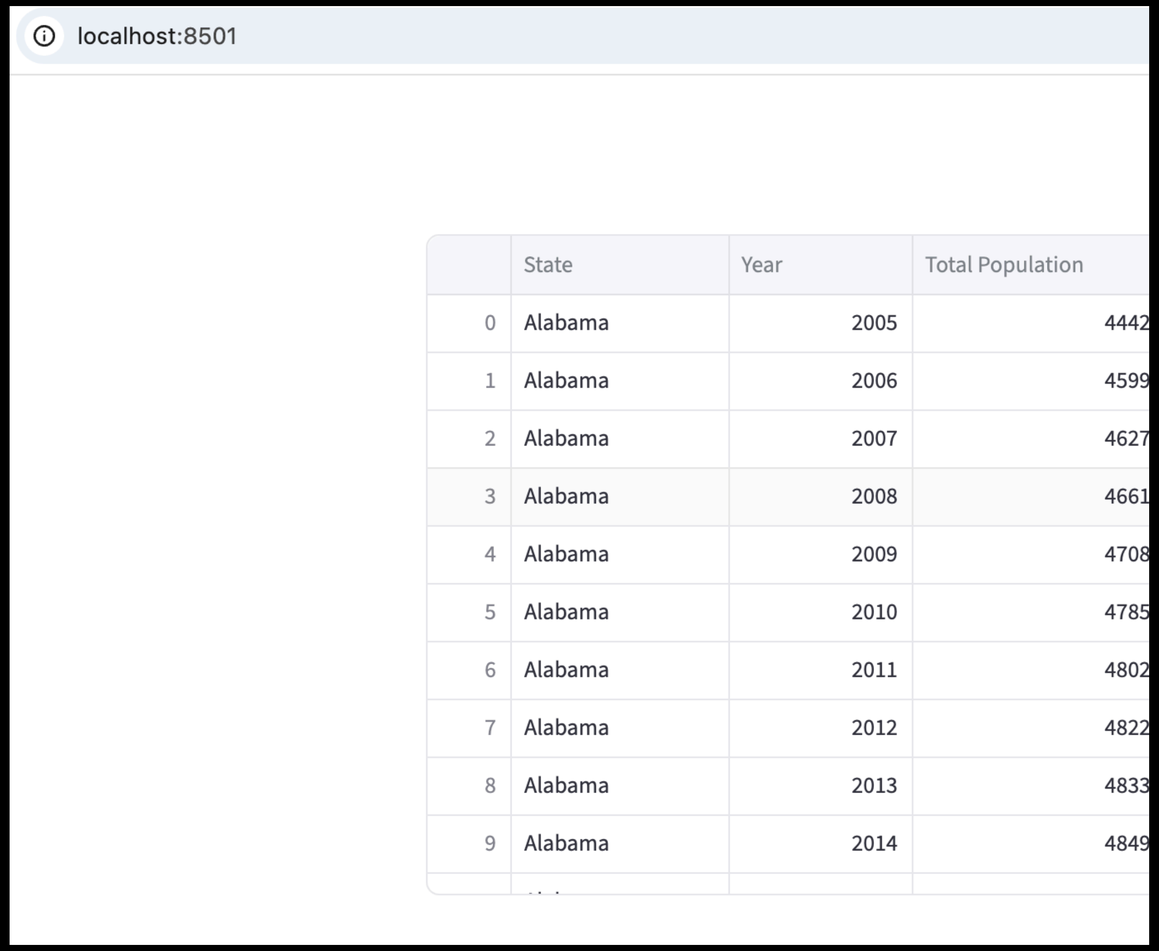
- Each session has its own directory
- Type `cd 1-intro-setup`
  - This moves to the directory for this session
- Type `streamlit run streamlit_app.py`
  - This runs the demo app for this section

```
(streamlit-workshop) ➜  streamlit-workshop git:(main) ✗ cd 1-intro-setup
(streamlit-workshop) ➜  1-intro-setup git:(main) ✗ streamlit run streamlit_app.py

  You can now view your Streamlit app in your browser.
```

Running the App

- Type streamlit run streamlit_app.py
- It should open a browser like this
- Click the table headers to sort (e.g. State, Year, Total Population)
- **Quiz:**
  - What are the min and max years of the dataset?
  - What state had the lowest population?
  - What state had the highest income?



	State	Year	Total Population
0	Alabama	2005	4442
1	Alabama	2006	4599
2	Alabama	2007	4627
3	Alabama	2008	4661
4	Alabama	2009	4708
5	Alabama	2010	4785
6	Alabama	2011	4802
7	Alabama	2012	4822
8	Alabama	2013	4833
9	Alabama	2014	4849

Recap: Setup

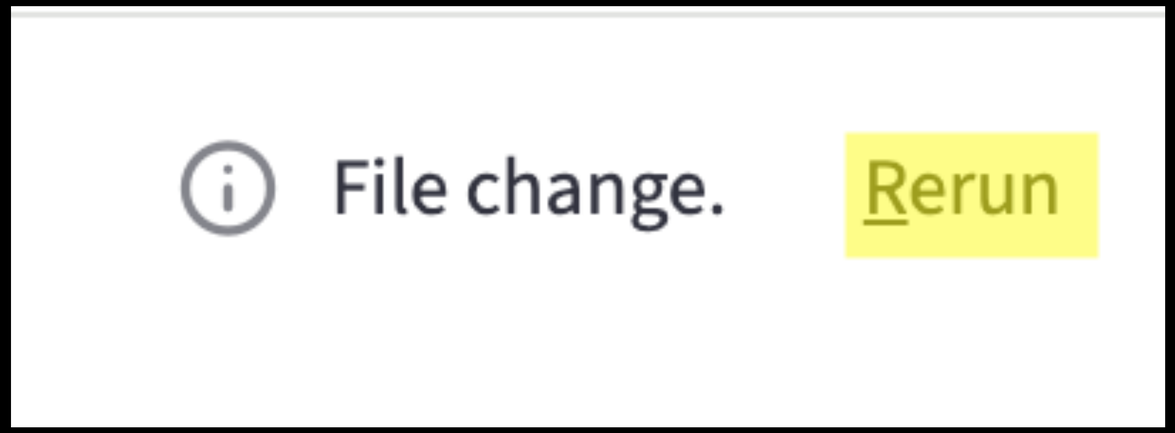
1. One-Time Setup
    1. Clone course repo
    2. Create the virtual environment
2. Per-Session Setup
    1. Activate the virtual environment
    2. Run the example Streamlit app

Course Agenda
(Big Picture)

1. Session 1: Setup & Basics

Rerun

- Streamlit is designed for fast iteration
- Each time you modify and save the script the word Rerun will appear in the top right of the app
- Click it to rerun the app with your changes
- Do not stop and restart the app from the command line



Exercise: Rerun

- In groups of 2
- Open the file streamlit_app.py in the directory 1-intro-setup
- Type streamlit run streamlit_app.py to run the app
- Uncomment the st.title line
- Save the file
- Click "rerun" and verify that the title appears

session1 > streamlit_app.py > ...

```python
import streamlit as st
import pandas as pd

#st.title("Demo Streamlit App")

df = pd.read_csv("state_data.csv")

st.dataframe(df)
```

Exercise: Rerun

- Result

Demo Streamlit App

	State	Year	Total Population	Median Household Income
0	Alabama	2005	4442558	36879
1	Alabama	2006	4599030	38783
2	Alabama	2007	4627851	40554
3	Alabama	2008	4661900	42666
4	Alabama	2009	4708708	40489
5	Alabama	2010	4785298	40474
6	Alabama	2011	4802740	41415
7	Alabama	2012	4822023	41574
8	Alabama	2013	4833722	42849
9	Alabama	2014	4849377	42830

Demo App Walkthrough

The data

- state_data.csv
- Basic demographic data of US States
- Each state, 2005-2023
- Comes from the American Community Survey (ACS)

session1 > state_data.csv > data

```csv
State,Year,Total Population,Median Household Income
Alabama,2005,4442558,36879
Alabama,2006,4599030,38783
Alabama,2007,4627851,40554
Alabama,2008,4661900,42666
Alabama,2009,4708708,40489
Alabama,2010,4785298,40474
Alabama,2011,4802740,41415
Alabama,2012,4822023,41574
Alabama,2013,4833722,42849
Alabama,2014,4849377,42830
Alabama,2015,4858979,44765
Alabama,2016,4863300,46257
```

Pandas

- The app uses the Pandas library to read in the csv file.
- Pandas is the most popular library in Python for working with tabular data
- Pandas is typically imported as pd
- pd.read_csv returns a dataframe

session1 > streamlit_app.py > ...

```python
import streamlit as st
import pandas as pd

#st.title("Demo Streamlit App")

df = pd.read_csv("state_data.csv")

st.dataframe(df)
```

Streamlit

- Lets the script run as a webapp
- Cannot run in a Jupyter notebook - must be a regular .py file.
- Imported as st
- Two output functions:
  - st.title
  - st.dataframe
- Processed top to bottom.

session1 > streamlit_app.py > ...

```python
import streamlit as st
import pandas as pd

st.title("Demo Streamlit App")

df = pd.read_csv("state_data.csv")

st.dataframe(df)
```

Demo Streamlit App

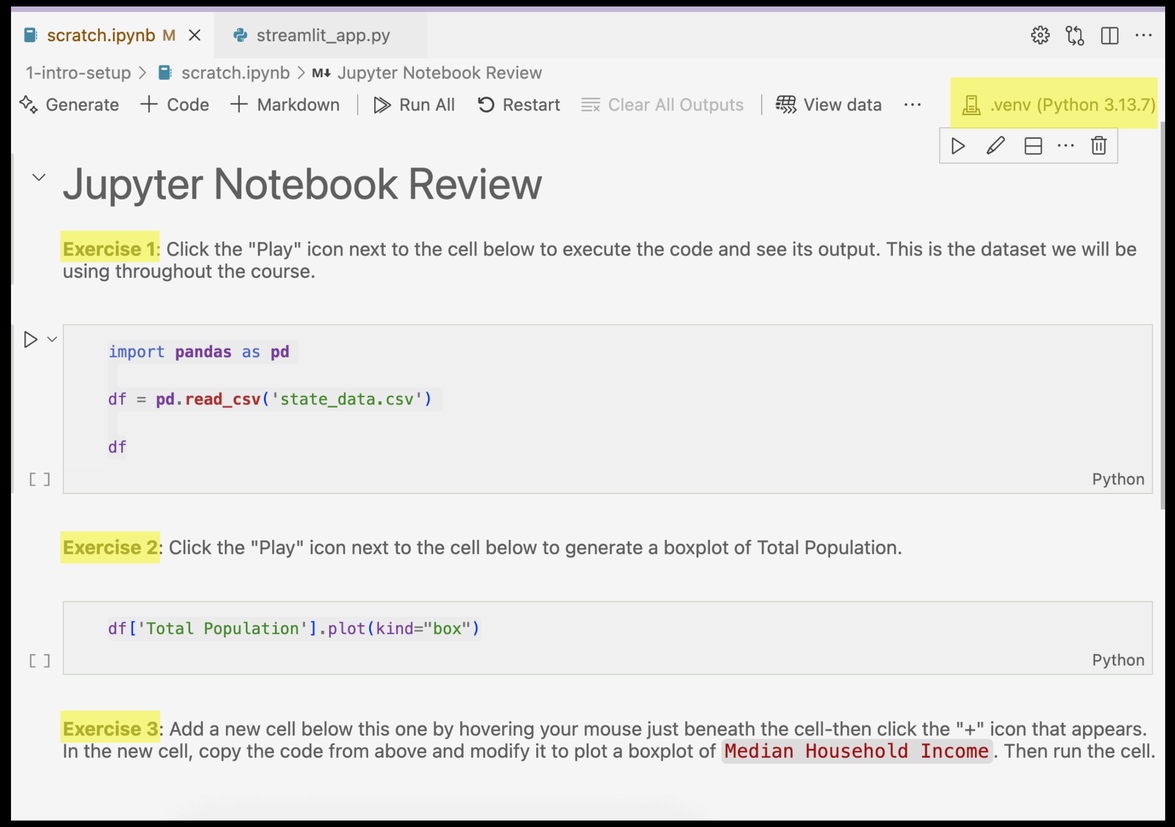
	State	Year	Total Population	Median Household Income
0	Alabama	2005	4442558	36879
1	Alabama	2006	4599030	38783
2	Alabama	2007	4627851	40554
3	Alabama	2008	4661900	42666

Exercise: Jupyter Notebooks

- Open scratch.ipynb. Two ways:
  1. VS Code with Jupyter plugin
  2. Jupyter notebook via command line:

     uv run jupyter notebook scratch.ipynb

- Complete the 3 exercises as a review



Exercise: LLM

- In groups of 2
- Tell an LLM
  - What you've done with Streamlit so far
  - What you should try next
- Experiment with one of the functions it recommends.
  - Use "rerun" to see the changes
- I use copilot.microsoft.com, but I think any LLM is fine.
- Stuck? Ask it for help!

I just started using Streamlit. So far I've used two functions: st.title and st.dataframe. Please recommend a few other simple, common functions I can use on a demo app.

Nice start, Ari! Since you've already used `st.title()` and `st.dataframe()`, here are a few other beginner-friendly Streamlit functions that will make your demo app more interactive and polished:

🖼️ Presentation & Layout

These help structure your app and guide the viewer:

- `st.header("Section Title")` — Adds a bold section header
- `st.markdown("**Bold text**, _italic_, [link](https://example.com)")` — Rich formatting
- `st.write("Any object")` — Smart display for text, data, plots, etc.
- `st.image("path_or_url")` — Show an image (e.g. logo or chart)

st.markdown

- Official docs
- Google "st.markdown"
- It can print emojis?!

Search

st.markdown

Version 1.49.0

Display string formatted as Markdown.

Function signature [source]

```
st.markdown(body, unsafe_allow_html=False, *, help=None, width="stretch")
```

Parameters

body *(any)*

The text to display as GitHub-flavored Markdown. Syntax information can be found at: https://github.github.com/gfm. If anything other than a string is passed, it will be converted into a string behind the scenes using `str(body)`.

This also supports:

- Emoji shortcodes, such as `:+1:` and `:sunglasses:`. For a list of all supported codes, see https://share.streamlit.io/streamlit/emoji-shortcodes.
- Streamlit logo shortcode. Use `:streamlit:` to add a little Streamlit flair to your text.
- A limited set of typographical symbols. `"<- -> <-> -- >= <= ~="` becomes "← → ↔ — ≥ ≤ ≈" when parsed as Markdown.

```
st.markdown("**I** :heart: *Streamlit*!")
```

Demo Streamlit App

I ❤️ *Streamlit*!

	State	Year	Total Population
0	Alabama	2005	4

Section 1 Review

- What is a data app?
- 1-time setup:
  - Clone course repo
  - Create virtual environment
- Per-session setup:
  - Activate virtual environment
  - Run app
- Rerun app without restarting Streamlit
- Review technical basics: Pandas, Jupyter Notebooks, Streamlit
- Ask LLM for a suggestion and implemented it